

Siddhant Poudyal

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EDUCATION

- **Indian Institute of Science, Education and Research** Mohali, India
Integrated BS-MS in Physics; CPI: 8.00 August 2019 - June 2024

PUBLICATIONS

- V. Satarifard, L. Sisson, Y. Han, ..., J. D. Mainland, P. Meyer, and the DREAM Olfactory Mixtures Prediction Consortium, including **Siddhant Poudyal**. A semantic-based community model for high-fidelity tuning of olfactory mixture distances. *PNAS*, 2026.
- Gautam Ahuja, Ayush Agarwal, Hara Prasad Mishra, Samagra Agrawal, Rik Ganguly, Zonunmawia, Akshay Sharma, Vatsal Batra, Bableen Kaur, **Siddhant Poudyal**, and 10 more authors. PRISM: Physician Rules Integrated with Small large language Models for probable diagnoses associated with Abdominal Pain. *NeurIPS 2025 Workshop on GenAI for Health*.

EXPERIENCE

- **Indian Institute of Technology, Roorkee**
Project Assistant, Hazra Lab Jul 2026 - Present
 - **NeurIPS AMP Challenge submission:** Implemented a generative design and ranking pipeline (VAE with conditional latent diffusion and a censoring-aware MIC ranker) for broad-spectrum antimicrobial peptides trained on a harmonized MIC dataset with CD-HIT homology-aware splits. [Link to repository](#)
 - **AMP/non-AMP classifier:** Trained an XGBoost classifier on ProtBERT embeddings (reduced using PCA, with collinear biophysical features removed) using homology-aware splitting, reaching a test MCC of 0.87 on a 156-peptide dataset. [Link to repository](#)
- **Ashoka University**
Project Research Scientist I, Augmented Health Systems Lab Jan 2026 - Jul 2026
 - Deployed a custom data workflow to harmonize over 40 highly unstructured clinical, socio-economic, and lab-based IVF-EHR data fields via rule-based extraction and manual curation
 - Developed an automated rule-based labeling and classification pipeline using 15+ lab parameters and EHR fields (Chief complaint, Past medical history, Family history, Current medication) to segment patients into clinical cohorts
- **Koita Centre of Digital Health - Ashoka University**
Machine Learning Engineer, Augmented Health Systems Lab Aug 2025 - Dec 2025
 - Built pipelines for docking and containerized deep learning models (DynamicBind, FlowDock, DiffDock) for Protein-Ligand docking and binding affinity prediction to identify candidate small-molecule inhibitors of the NS3 serine protease.
 - Deployed LLM-augmented pipelines for clinical term extraction from over 70,000 unstructured clinical prescription notes.
 - Implemented a semi-automated data pipeline to standardize unstructured EHR records (Chief complaint, Past medical history, Family history, Current medication) using OHDSI standardized vocabularies
 - Worked with subject matter expert to develop a pipeline for Somatic Variant Calling with tumor-only and paired tumor-normal mode on 15 samples
- **Koita Centre of Digital Health - Ashoka University**
Junior Machine Learning Engineer, Augmented Health Systems Lab Jul 2024 - Jul 2025
 - Designed and implemented web-scraping pipelines to scrape Indian medication data (both prescription and over-the-counter drugs) from popular e-pharma websites
 - Built a (semi-automated) workflow to map Indian medication data to OMOP at the Active Pharmaceutical Ingredient (API) Level.
 - Used the now-interoperable Indian medication data to find gaps in Drug Codes at the national level (NRCeS CDCI). Also linked the standardized medication data to global databases such as PubChem and DrugBank.

CONFERENCES, POSTERS AND TALKS

- **OHDSI India Symposium 2025:** Delivered talk on knowledge base specific to Indian medications and structured extraction using LLMs
- **Ashoka 2nd Centres Expo 2025:** Poster presentation of India-specific medication Knowledge base and clinical term extraction pipeline

HONORS AND COMPETITIONS

- DST-INSPIRE Scholar (2019-2024)
- Gold Medal at iGEM Jamboree 2022 (iGEM Team: IISER_Mohali)

SKILLS

- **Machine Learning:** PyTorch, scikit-learn, NumPy, XGBoost, Random Forests, VAEs, latent diffusion
- **Computational Biology:** Homology-aware data splitting (CD-HIT), protein-ligand docking (DiffDock, FlowDock, DynamicBind), ColabFold, GATK somatic variant calling
- **Data Engineering:** pandas, OMOP CDM, Knowledge graphs, NER pipelines, web scraping